A HIGH POWER OF AN IDEAL MAPS THROUGH THE DERIVED TENSOR PRODUCT OF THE IDEAL WITH THE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 29736-29742 (canonical tag 0928) and the counted proof is the proof environment at source lines 29744-29768, which immediately follows it and is its own proof.

Editorial note. The printed text writes It can also been seen directly where It can also be seen is meant; the wording is copied exactly as printed and is not corrected.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note. The counted statement and proof cite no bibliography entry, so this card needs no bibliography.

Allowed standard prerequisites (editorial).

- For an ideal in a Noetherian ring the comparison between the derived tensor product with its powers and the ordinary products is a pro-isomorphism, which immediately gives a power factoring through the derived tensor product of the ideal itself. invoked at proof lines 29744-29745. Supplies a short proof of the statement, after which the author gives an independent direct argument.
- The quotient by the ideal sits in a distinguished triangle with the ideal, so factoring through the derived tensor product with the ideal is equivalent to the vanishing of the composite into the derived tensor product with the quotient. used at proof lines 29746-29752. Reformulates the goal as the vanishing of one morphism in the derived category.
- A finite module annihilated by an ideal can be absorbed by a sufficiently high power of that ideal in the Ext group, so an obstruction element can be killed after increasing the power. invoked at proof lines 29762-29765. Allows the successive factorization steps to be completed in finitely many rounds, which gives the required power.

Source reference. Original lemma 0G3N: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29677–29677. The reference is cited at proof line 29744 as the reason the statement holds, after which the author gives an independent direct argument; the direct argument, which is what the counted proof contains, uses this reference only as a remark and it is the subject of a separate card of this lane.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0927: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29571–29571. The reference is applied at proof lines 29762-29765 to kill the obstruction element in the Ext group after increasing the power of the ideal; the induction that consumes it is inside the counted span.

Lemma 0.2. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. Let M be a finite A-module. There exists an integer $n > 0$ such that $I^n M \to M$ factors through the map $I \otimes_A^{\mathbf{L}} M \to M$ in $D(A)$.*

Proof. This follows from Lemma 0G3N. It can also been seen directly as follows. Consider the distinguished triangle

$$I \otimes_A^{\mathbf{L}} M \to M \to A/I \otimes_A^{\mathbf{L}} M \to I \otimes_A^{\mathbf{L}} M[1]$$

By the axioms of a triangulated category it suffices to prove that $I^n M \to A/I \otimes_A^{\mathbf{L}} M$ is zero in $D(A)$ for some n. Choose generators $f_1, \ldots, f_r$ of I and let $K = K_\bullet(A, f_1, \ldots, f_r)$ be the Koszul complex and consider the factorization $A \to K \to A/I$ of the quotient map. Then we see that it suffices to show that $I^n M \to K \otimes_A M$ is zero in $D(A)$ for some $n > 0$. Suppose that we have found an $n > 0$ such that $I^n M \to K \otimes_A M$ factors through $\tau_{\geq t}(K \otimes_A M)$ in $D(A)$. Then the obstruction to factoring through $\tau_{\geq t+1}(K \otimes_A M)$ is an element in $\mathrm{Ext}^t(I^n M, H_t(K \otimes_A M))$. The finite A-module $H_t(K \otimes_A M)$ is annihilated by I. Then by Lemma 0927 we can after increasing n assume this obstruction element is zero. Repeating this a finite number of times we find n such that $I^n M \to K \otimes_A M$ factors through $0 = \tau_{\geq r+1}(K \otimes_A M)$ in $D(A)$ and we win. □